

Laboratory of Applied & Industrial Electronics

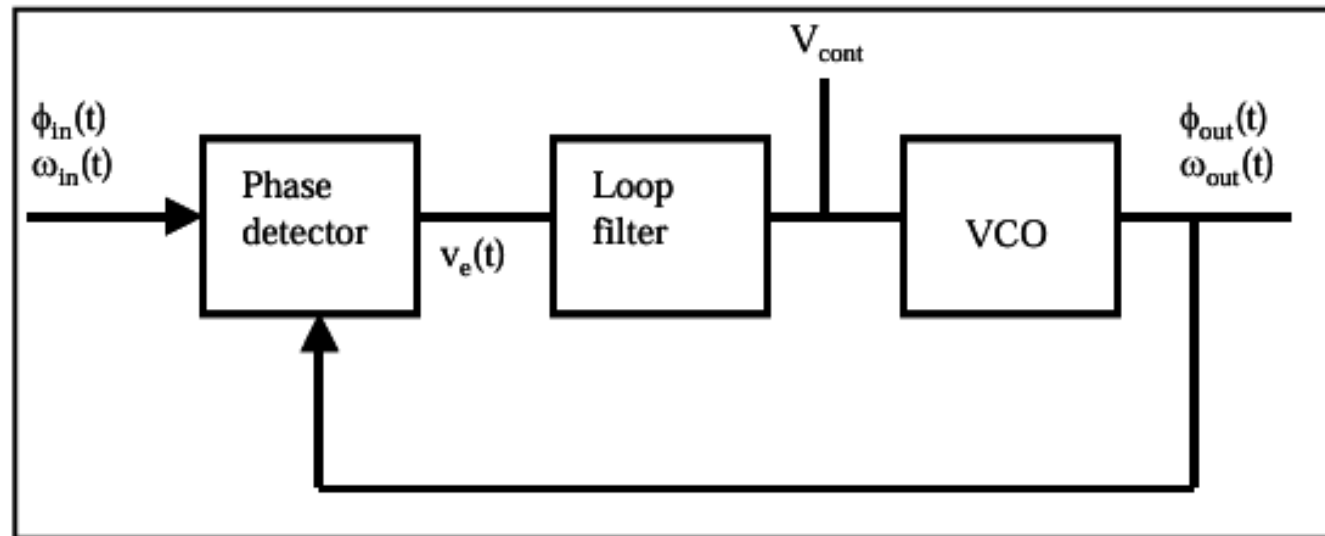
#4 PLL

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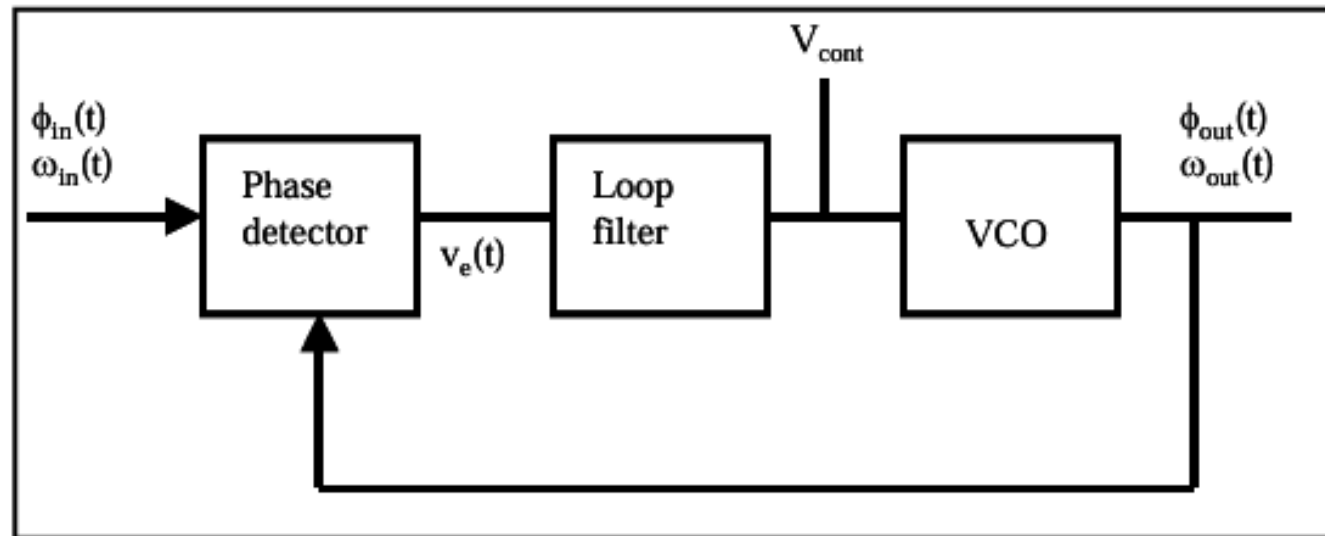
Introduction

- A PLL is a feedback system that includes a voltage controlled oscillator, phase detector, and lowpass filter within its loop.
- Its purpose is to force the VCO to replicate and track the frequency and phase at the input when in lock.
- It is possible to have a phase offset between input and output, but when locked, the frequencies must exactly track.



Introduction

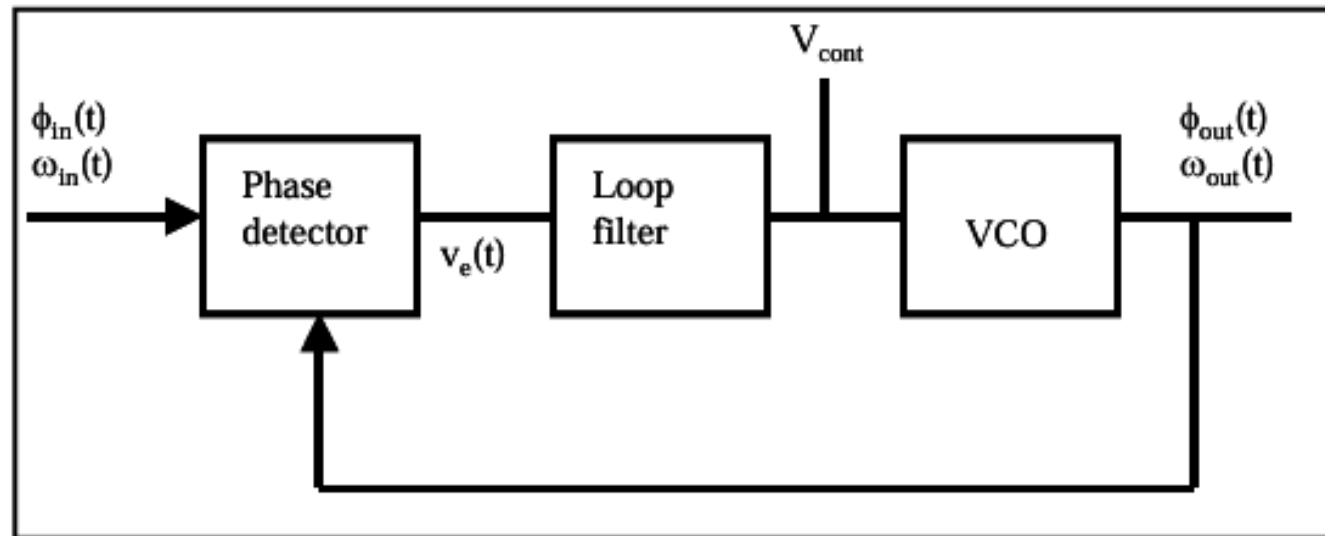
- The PLL output can be taken from either V_{cont} , the filtered (almost DC) VCO control voltage, or from the output of the VCO depending on the application.
- The former provides a baseband output that tracks the phase variation at the input.
- The VCO output can be used as a local oscillator or to generate a clock signal for a digital system.
- Either phase or frequency can be used as the input or output variables.



Phase detector

- The phase detector compares the phase at each input and generates an error signal, $v_e(t)$, proportional to the phase difference between the two inputs. K_D is the gain of the phase detector (V/rad):

$$v_e(t) = K_D[\phi_{out}(t) - \phi_{in}(t)]$$



Phase detector

- An analog multiplier or mixer can be used as a phase detector, let consider two sinusoidal signal $A(t)$ and $B(t)$ that are multiplied together:

$$\begin{aligned} v_{e(t)} &= A \cos(\omega_0 t + \phi_A) B \cos(\omega_0 t + \phi_B) = \\ &= \frac{AB}{2} [\cos(2\omega_0 t + \phi_A + \phi_B) + \cos(\phi_A - \phi_B)] \end{aligned}$$

- Since the two inputs are at the same frequency when the loop is locked, we have one output at twice the input frequency and an output proportional to the cosine of the phase difference.
- The doubled frequency component must be removed by the lowpass loop filter.
- Any phase difference then shows up as the control voltage to the VCO, a DC or slowly varying AC signal after filtering.

Voltage controlled oscillator

- In PLL applications, the VCO is treated as a linear, time-invariant system.
- Excess phase of the VCO is the system output is

$$\phi_{out} = K_O \int_{-\infty}^t V_{cont} dt'$$

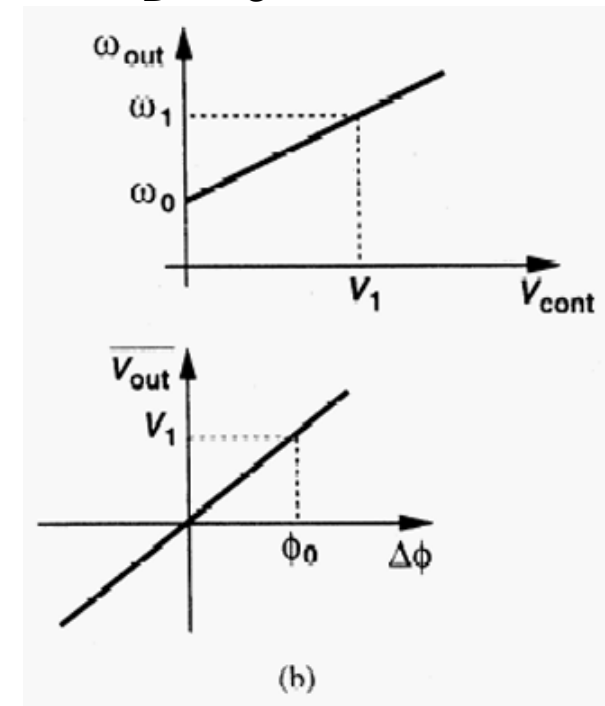
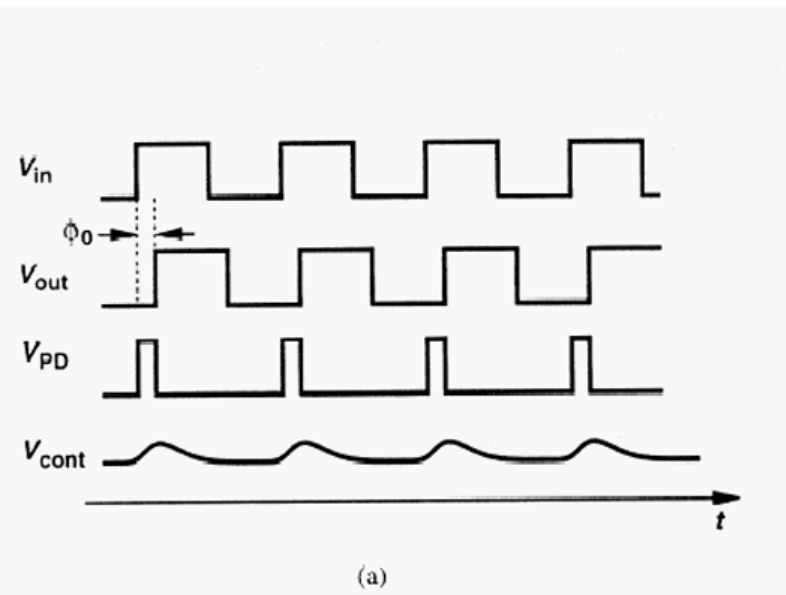
- The VCO oscillates at an angular frequency, ω_{out} . Its frequency is set to a nominal ω_0 when the control voltage is zero.
- Frequency is assumed to be linearly proportional to the control voltage with a gain coefficient K_O or K_{VCO} (rad/s/v).

$$\omega_{out} = \omega_0 + K_O V_{cont}$$

- Thus, to obtain an arbitrary output frequency (within the VCO tuning range, of course), a finite V_{cont} is required. Let's define $\phi_{out} - \phi_{in} = \phi_0$.

Voltage controlled oscillator

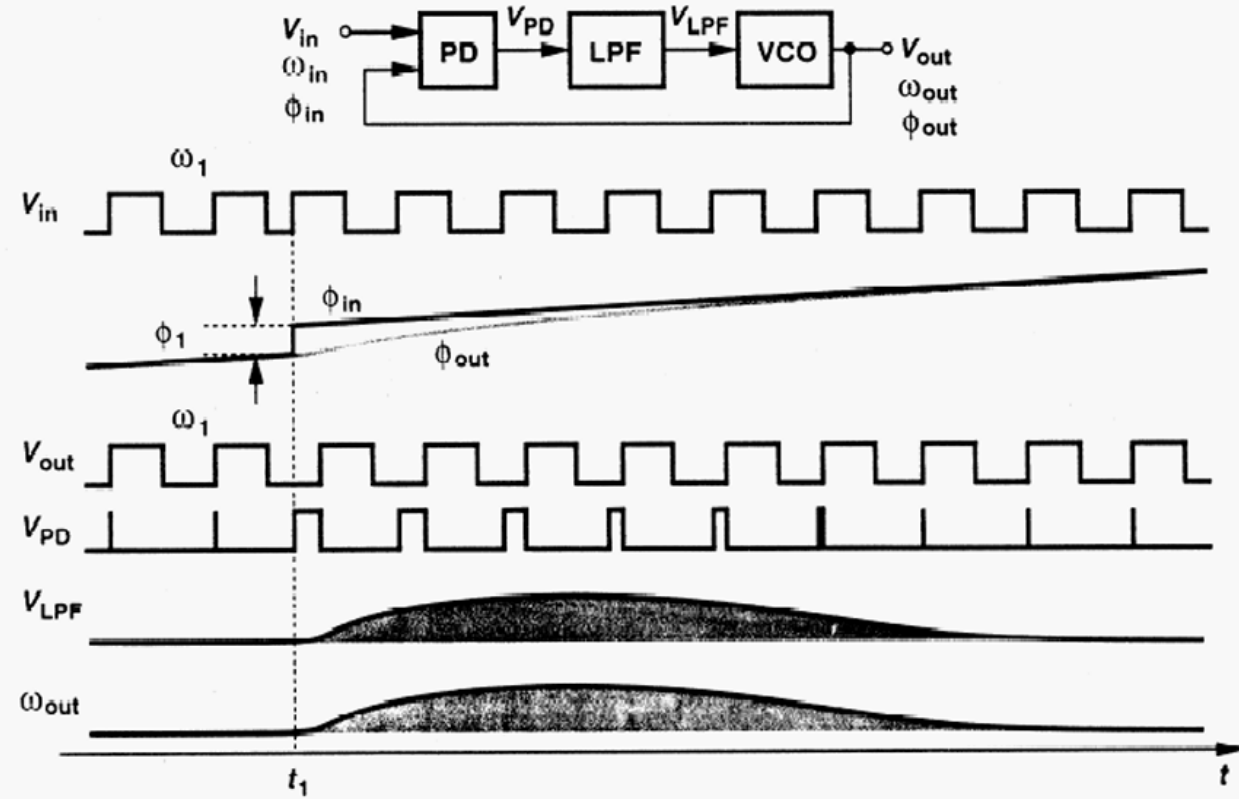
- Let's define $\phi_{out} - \phi_{in} = \phi_0$. In the figure, the two inputs to the phase detector are depicted as square waves that are put in XOR to produce an output pulse whenever there is a phase misalignment.
- Suppose that an output frequency ω_1 is needed. From the right figure, we see that a control voltage V_1 will be necessary to produce this output frequency.
- The phase detector can produce this voltage only by maintaining a phase offset ϕ_0 at its input. To minimize the required phase offset or error, the PLL loop gain, K_D, K_O , should be maximized, since : $\phi_0 = \frac{V_1}{K_D} = \frac{\omega_1 - \omega_0}{K_D K_O}$



PLL dynamic response

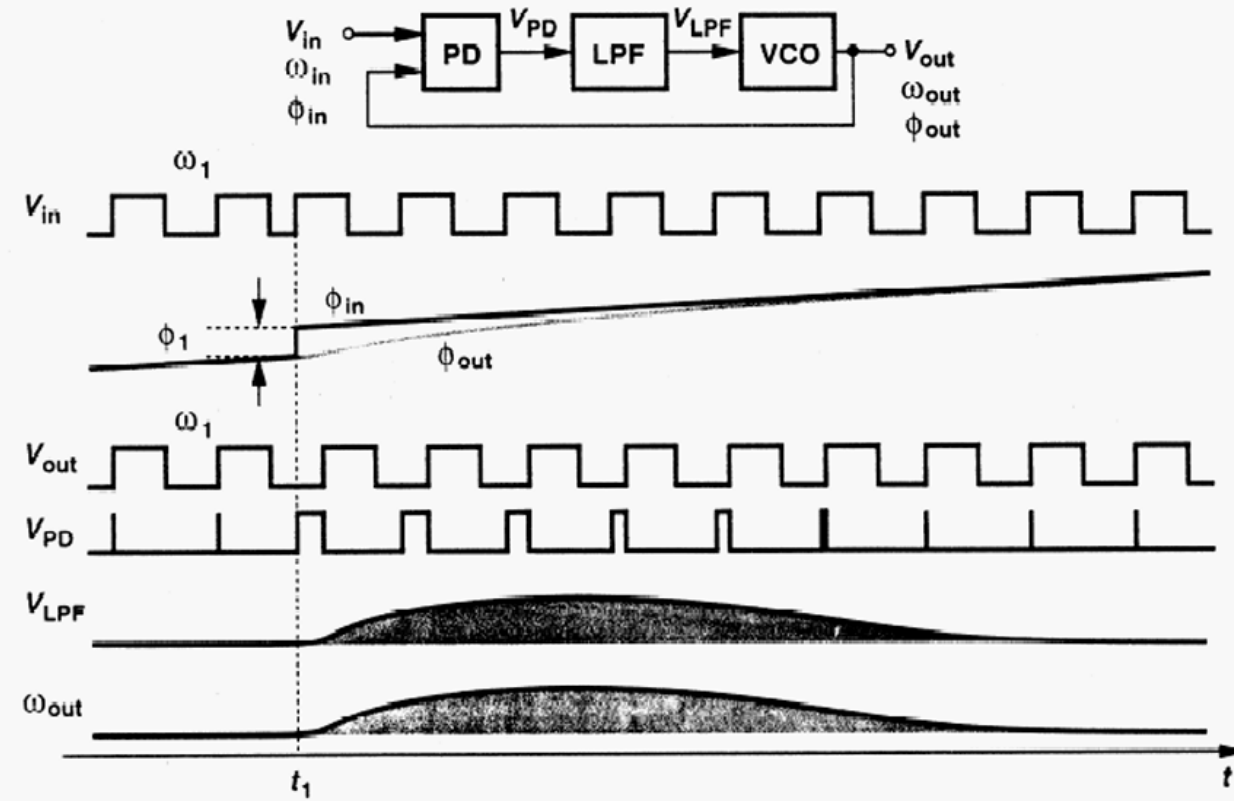
- To see how the PLL works, suppose that we introduce a phase step at the input at $t = t_1$: $\phi_{in} = \omega_1 t + \phi_0 + \phi_1 u(t - t_1)$
- Since we have a step in phase, the initial and final frequencies must be identical: ω_1 .
- A temporary change in frequency is necessary to shift the phase by ϕ_1 .
- The area under ω_{out} gives the additional phase because V_{cont} is proportional to frequency.

$$\phi_1 = \int_{t_1}^{\infty} \omega_{out} dt = \int_{t_1}^{\infty} K_O V_{cont}(t) dt$$



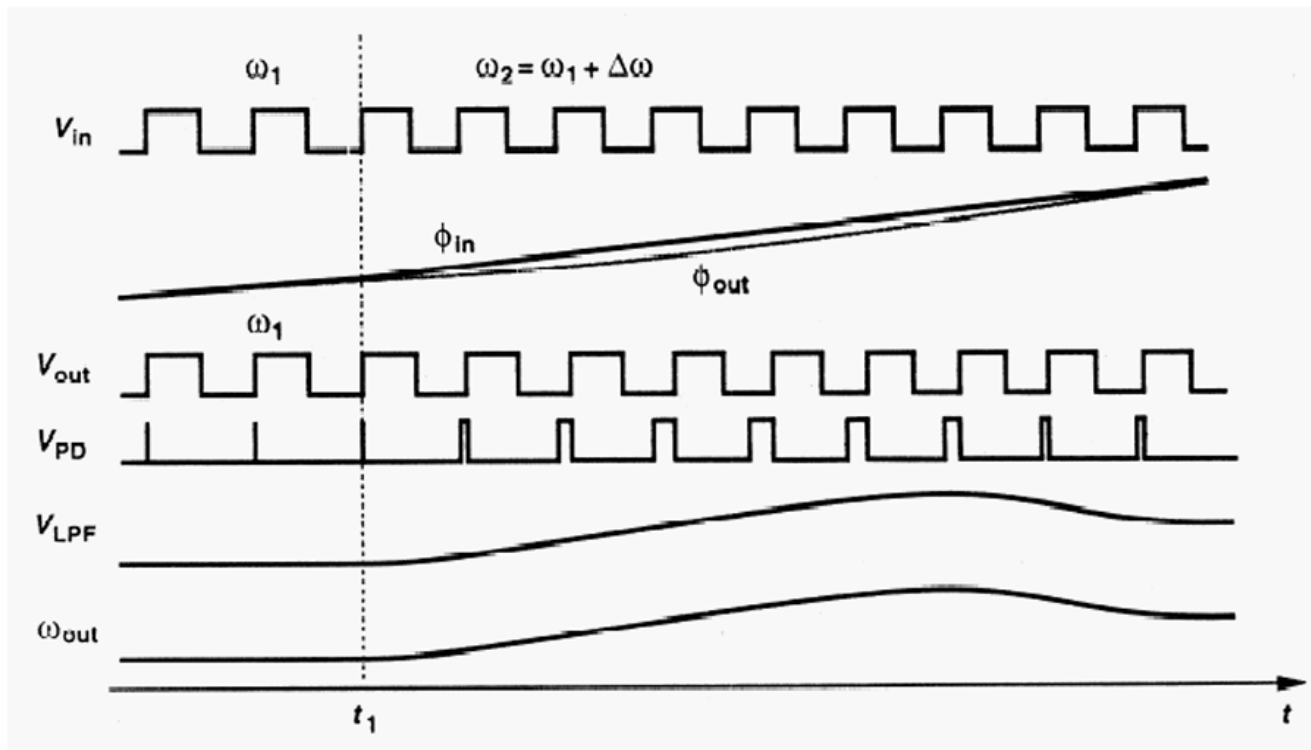
PLL dynamic response

- After settling, all parameters are as before since the initial and final frequencies are the same.
- This shows that $V_{cont}(t)$ [shown as $V_{LPF}(t)$ in the figure] can be used to monitor the dynamic phase response of the PLL.



PLL dynamic response

- Now, let's investigate the behavior during a frequency step: $\omega_2 = \omega_1 + \Delta\omega$
- The frequency step will cause the phase difference to grow with time since a frequency step is a phase ramp. This in turn causes the control voltage, V_{cont} , to increase, moving the VCO frequency up to catch up with the input reference signal.
- In this case, we have a permanent change in ω_{out} since a higher V_{cont} is required to sustain a higher ω_{out} .
- If the frequency step is too large, the PLL will lose lock.



Lock range and capture range

- The lock range is the range of input signal frequencies over which the loop remains locked once it has captured the input signal. This can be limited either by the phase detector or the VCO frequency range.
- If it is limited by phase detector, the range depends on DC loop gain and not on the loop filter gain.
- The lock range could also be limited by the tuning range of the VCO. Oscillator tuning range is limited by capacitance ratios or current ratios and is finite. In many cases, the VCO can set the maximum lock range.
- The capture range is the range of input frequencies around the VCO center frequency onto which the loop will lock when starting from an unlocked condition.
- Sometimes a frequency detector is added to the phase detector to assist in initial acquisition of lock.